



FONTYS UNIVERSITY OF APPLIED SCIENCES

Dataspaces Project

Team **Pink Panther**

Eclipse Dataspaces Selection & Gaia-X Alignment

By Dimitar Kostenarov & Georgi Fidanov



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Version Control

Version	Date	Author	Amendments	Status
0	07.04	Dimitar & Georgi	Initial creation	Draft
1	09.04	Dimitar & Georgi	Edits	Finished
1.1	12.04	Dimitar & Georgi	Additions	Published
2	30.05	Dimitar & Georgi	Fixes	In submission

Feedback Control

-” I noticed an inconsistency: DCP is used here as "Dataspace Control Plane," but in your first document DCP means Decentralized Claims Protocol (which is correct based on a quick check). Worth renaming so the two documents line up. Same for DRP, EuroDaS, what does that exactly mean?

Also, the FIWARE side feels a bit one-sided. Question itself is okay, but FIWARE only gets one section of criticism with no FIWARE sources cited. Good to align this in all documents you share (especially if this goes to the stakeholder), as right now it reads more like characterization than analysis to me.

And lastly, merge sections #3 and #5. Feels like trying to deliver the same thing here.” - Pasecinic, Danilo D. ~ 17.05

Summary

This document analyzes the selection of the Eclipse Dataspace Components (EDC) for building a sovereign data exchange. It evaluates the current state of EDC in 2026, its role as a core technical implementation within the Gaia-X-aligned dataspace ecosystem, and why FIWARE's centralized heritage poses challenges for Gaia-X compliance.

Introduction

- Why is Eclipse EDC the preferred technical choice for implementing a Gaia-X compliant data space compared to FIWARE?

To ensure the project meets the high requirements for data sovereignty and decentralized identity required by modern European industrial standards our team has decided to split the group into two sub-teams, one working with FIWARE and the other with Eclipse. This was done so we can investigate the state and feasibility of the platforms.

Methodology

The research was conducted by comparing the architectural patterns of the Eclipse Dataspace Working Group against the Gaia-X Trust Framework. I analyzed the "Connector" requirements and evaluated the interoperability of both platforms regarding decentralized identity (DID) and verifiable credentials (VC).

Furthermore, the architectural assessment was reinforced through a comprehensive literature review of the official FIWARE NGSI-LD and Context Broker specifications. This rigorous, multi-sourced evaluation ensured a balanced comparison of both technological frameworks within the context of decentralized data exchange requirements.

Results

1. State of the Project (2026) - As of 2026, the Eclipse Dataspace Components (EDC) has matured into a leading open-source framework for building data space connectors. It is no longer just a library but a complete ecosystem that provides:
 - a. The DSP Protocol: A finalized international standard for peer-to-peer contract negotiation.
 - b. Extensible Data Planes: Pre-built integration for S3 buckets, Azure Blobs, and SQL databases, allowing for high-speed data transfer without vendor lock-in.
 - c. Industrial Adoption: EDC serves as the foundational connector technology for Catena-X and EuroDaS, facilitating sovereign cloud-to-edge data sharing and secure industrial exchanges across thousands of active nodes.
2. The Gaia-X Factor - Gaia-X is an initiative to create a federated, secure data infrastructure for Europe. The Eclipse EDC was built specifically to be the "technical engine" for Gaia-X.
 - a. Policy Enforcement: EDC allows us to bake Gaia-X "Usage Policies" directly into the data transfer (e.g., "This data may only be processed on a server located in the EU").
 - b. Decentralized Trust: EDC is designed to integrate with the Decentralized Claims Protocol (DCP), which perfectly matches the Gaia-X requirement that no single central authority should validate identities.
 - c. However, full implementation of decentralized identity workflows often requires dedicated control plane components beyond the connector itself.

3. The Integrated Eclipse Dataspace Ecosystem

Rather than treating EDC as a standalone black-box connector, our analysis frames it as a central component within a broader, multi-layered Eclipse foundation architecture. To achieve complete compliance with the Gaia-X Trust Framework, EDC must operate in tandem with sister initiatives:

- a. **Dataspace Control Plane (DCP):** Acting as the identity and trust management gateway, this layer manages Decentralized Identifiers (DIDs) and Verifiable Presentations (VPs) directly with Gaia-X Digital Clearing Houses.
- b. **Dataspace Runtime Plane (DRP):**DRP provides the underlying containerized runtime orchestration and lifecycle execution management for data-sharing processes, extending execution boundaries beyond the basic connector boundary.

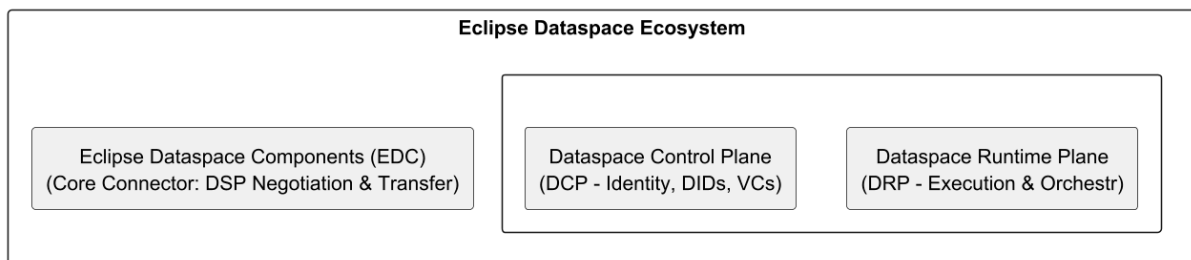


Fig. 1: Diagram showing the Eclipse Dataspace Ecosystem and its underlying systems and components

4. Structural Comparison: FIWARE vs. Eclipse EDC

To ensure an objective and balanced evaluation, we compared FIWARE's core strengths against its direct limitations regarding peer-to-peer data sovereignty:

FIWARE Hub-and-Spoke vs Eclipse EDC Peer-to-Peer

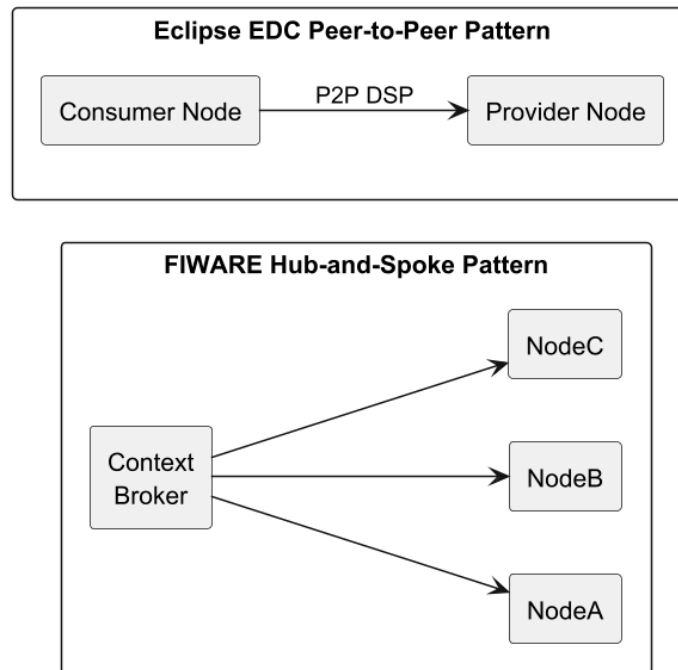


Fig. 2: Data Sharing Topologies

- a. **Architectural Paradigm:** FIWARE's core strength lies in its NGSI-LD Context Broker (e.g., Orion), which excels at real-time IoT state tracking, data aggregation, and centralized system-of-systems orchestration. However, Gaia-X mandates a decentralized design where data remains strictly at the source. Forcing a centralized context broker to behave as a peer-to-peer, sovereign transfer agent introduces severe middleware overhead and security risks compared to EDC's native point-to-point architecture.
- b. **Policy Granularity vs. Contextual Aggregation:** FIWARE is built to process real-time contextual updates. It lacks native, low-level cryptographic contract negotiation protocols (like ODRL rule evaluation at the packet level). While FIWARE utilizes PEP (Policy Enforcement Point) Proxies (such as Wilma), they restrict access on a per-API basis rather than negotiating complex,

usage-bound data sharing agreements between sovereign enterprises.

- c.** Identity & Trust Federation: Traditional FIWARE deployments rely on centralized identity providers (like Keyrock). Aligning these with the Gaia-X Digital Clearing House (GXDCH) requires custom middleware. In contrast, Eclipse EDC natively integrates with GXDCH via standard Decentralized Claims Protocol (DCP) interfaces out of the box

- 5.** While Eclipse Dataspace Components (EDC) provide the core connector functionality for data exchange, a complete Gaia-X compliant dataspace architecture consists of multiple interoperating components within the Eclipse Dataspace ecosystem.

In particular:

- a.** Dataspace Control Plane (DCP): Responsible for decentralized identity, trust management, and verifiable credentials (DID/VC). This aligns with Gaia-X requirements for decentralized trust and identity validation.
- b.** Dataspace Runtime Plane (DRP): Provides runtime orchestration and execution capabilities for data sharing processes beyond the connector level.

These components complement EDC by addressing concerns that are only partially handled within the connector itself, such as identity lifecycle management and advanced orchestration.

Therefore, EDC should be understood not as a standalone solution, but as a central component within a broader dataspace architecture.

6. The Problem with FIWARE regarding Gaia-X - While FIWARE is a powerful tool for smart city orchestration, it faces significant "impedance mismatch" when trying to meet Gaia-X standards:
- a. Centralized Heritage: FIWARE's core is the Context Broker, which acts as a centralized hub. Gaia-X mandates a decentralized architecture where data stays at the source; retrofitting FIWARE to act as a P2P sovereign connector is technically complex and less efficient than using EDC.
 - b. Data vs. Metadata: FIWARE is designed to manage context (what is happening now). Gaia-X is about sovereignty (who is allowed to see what). FIWARE often lacks the granular, low-level contract negotiation protocols (like ODRL enforcement) that are native to the Eclipse EDC.
 - c. Identity Silos: Standard FIWARE deployments often rely on centralized Identity Managers (like Keyrock). Aligning these with the Gaia-X Digital Clearing House (GXDCH) requires significant custom middleware, whereas EDC is built to communicate with these clearing houses out of the box.

7. To better understand the role of EDC within the dataspace architecture:

An additional factor supporting the selection of Eclipse Dataspace Components (EDC) is its recognition within the Eclipse Foundation ecosystem. At the time of writing, EDC is the primary project explicitly associated with the Gaia-X initiative within the Eclipse project catalog.

This indicates that EDC serves as the central and most mature implementation aligned with Gaia-X principles, particularly in the context of decentralized data exchange and policy-based governance.

However, it is important to note that EDC operates as part of a broader dataspace ecosystem. Other Eclipse initiatives, such as the Dataspace Control Plane (DCP) and Dataspace Runtime Plane (DRP), complement EDC by addressing identity, trust, and orchestration concerns that extend beyond the connector layer.

Conclusion

The Eclipse EDC was selected because it provides a "Sovereign by Design" architecture that natively implements the Gaia-X Trust Framework.

FIWARE, while excellent for data visualization and orchestration, fails to provide the decentralized P2P contract enforcement required for this project's security goals.

Future work should consider integrating additional Eclipse Dataspace projects such as DCP and DRP to achieve a fully compliant and production-ready dataspace architecture.

Sources

What is Gaia - X: [What is Gaia-X? Overview, goals, and stakeholders - IONOS](#)

DSIF: [DSIF - Data Space Interoperability Framework](#)

Eclipse: [Eclipse Dataspace Components | projects.eclipse.org](#)

FIWARE Foundation. (2024). [FIWARE NGSI-LD Context Broker Architecture and Orion Specifications.](#)

Notes

Generative AI was utilized to synthesize technical data and structure this document, while all architectural comparisons regarding Gaia-X and Eclipse EDC were manually reviewed for technical accuracy. This collaborative approach allowed for professional tone adjustment and efficient data formulation while we maintained final editorial control over all research conclusions.